Organism Discovery Report

This Organism Discovery Report (ODR) was generated by TaxTriage, a flexible, containerized bioinformatics pipeline for pathogen detection from untargeted DNA or RNA sequencing data. TaxTriage supports both short-read (Illumina) and long-read (ONT, PacBio) sequencing platforms and incorporates quality control, organism classification, read mapping, and a unified confidence metric for all detected organisms.

The ODR provides a consolidated summary of the evidence supporting pathogen identification, including key results from the intermediate analyses performed throughout the pipeline. TaxTriage is intended for broad deployment and early-stage outbreak investigations and is not intended as a standalone diagnostic capability.

Summary Statistics

4 Number of Samples	**158,494** Total Reads Analyzed	**89.4%** % of Reads Aligned
16 Unique Organisms	**26** Unique Genera	**6** High Consequence Pathogens Detected

Organism Discovery Report

TAXTRIAGE VERSION	GENERATED	COMMIT ID
Not specified / local build	2026-07-29 18:19	local

Color Key

	Category	Description
	Primary Pathogen (Matched Sample Type)	Detected by taxonomic id or assembly name and classified as clinically relevant for this sample type.
	Primary Pathogen (Other Sample Type)	Detected and classified as clinically relevant for a different sample type.
	Species/Subkey Summary (inferred)	Species-level detection exceeds the threshold, but individual strains do not. (!) indicates the highest scoring primary strain below the threshold.
	Potential	Potential pathogen requiring further investigation
■	DNA Pathogen	Detected pathogen has a DNA genome.
■	RNA Pathogen	Detected pathogen has an RNA genome.
★	High Consequence	Organism of elevated clinical significance that may warrant immediate review.
	[site flora]	Identifies organisms that are normal flora at the indicated body site but were detected in a normally sterile sample (blood, CSF, or serum).

Categories excluded by report settings and omitted from this report: **Commensal, Opportunistic, Unknown**. See the output reports or multi-run analysis for additional details.

Column Explanations

Organism	Detected organism with associated annotation, taxonomic ID, and taxonomic rank.
TASS Score	Confidence score for organism detection (0–100), with higher values indicating greater confidence.
Classifier Reads	Number of reads assigned to the organism by Kraken2/Centrifuge.
Aligned Reads	Number of reads aligned to the organism's reference genome. (%) indicates the proportion of total sample reads aligned to that organism. **Aligned read counts may differ from the original mapping results due to post-processing steps that remove potential false-positive alignments.**
RPM	Reads Per Million (RPM), a normalized abundance metric that enables comparison across samples.
% Coverage	Percentage of the organism's genome covered by aligned reads.
High ANI Matches	Lists strains with Average Nucleotide Identity (ANI) above the specified threshold.
Control Comparison	Displays an organism's TASS score relative to control samples. • **Red**: Negative-control range. • **Green**: Positive-control range. • **Purple**: In-silico simulation range, when available. • **×**: Sample TASS fold change; **× rd**: Read-count fold change. • Prefix symbols: – max negative control, **+** min positive control, ∞ in-silico control.

Run Summary

Sample	Sample Type	Platform	Total Reads	Aligned Reads	Organism Reads
Miseq_Run_A	nasal	illumina	117,322	105,423	105,423
Nanopore_Run_A	stool	oxford	3,000	2,541	2,541
Negative_Control_Miseq	nasal	illumina	172	144	144
Positive_Control_Miseq	nasal	illumina	38,000	33,614	33,614

Per-Sample Classification Results

Sample 1: Miseq_Run_A

SAMPLE TYPE	SEQUENCING PLATFORM	TOTAL READS	ALIGNED READS	SPECIES GROUPS	TASS CUTOFF
nasal	illumina	117,322	105,423	9	75.0

Organism	TASS Score	Classifier Reads	Aligned Reads	RPM	% Coverage	Control Comparison
Orthopneumovirus *1 species; 2 qualifying strains*						
Orthopneumovirus hominis ■ **(3049954)** *species*	100.0	76	251 (0.2%)	1,621	100.0%	+ 1.0×5.0× rd
! Human respiratory syncytial virus A ■ (208893) *qualifying strain*	100.0	51	201 (0.2%)	1,907	100.0%	+ 1.0×4.0× rd
! human respiratory syncytial virus (B) ■ (11250) *qualifying strain*	92.5	76	50 (0.0%)	474	58.2%	+ 0.9×1.0× rd
Betacoronavirus *1 species; 1 qualifying strain*						
Severe acute respiratory syndrome coronavirus 2 ★ ■ **(2697049)** *strain*	100.0	49	98 (0.1%)	930	63.6%	-
Pseudomonas *1 species; 1 qualifying strain*						
Pseudomonas aeruginosa ■ **(287)** *species*	99.9	20,693	40,341 (34.4%)	379,282	74.6%	+ 2.1×200.7× rd – 4.0×2881.5× rd
Orthopoxvirus *1 species; 1 qualifying strain*						
Monkeypox virus ★ ■ **(10244)** *strain*	98.6	650	1,218 (1.0%)	11,553	77.4%	+ 1.2×152.2× rd
Neisseria *1 species; 1 qualifying strain*						
Neisseria gonorrhoeae ■ **(485)** *species*	99.0	3,575	7,145 (6.1%)	63,961	59.3%	+ 1.0×1.0× rd – 4.5×893.1× rd
Listeria *1 species; 1 qualifying strain*						
Listeria monocytogenes ★ ■ **(1639)** *species*	97.8	4,856	8,607 (7.3%)	79,821	50.8%	+ 2.4×100.1× rd – 4.7×1434.5× rd

Positive Control Organisms Not Detected in Sample: *Enterococcus* (1350) Control TASS 40.89, 81 reads, source: Positive_Control_Miseq.paths.json; *Limosilactobacillus* (2742598) Control TASS 90.03, 70 reads, source: Positive_Control_Miseq.paths.json; *Salmonella* (590) Control TASS 93.09, 193 reads, source: Positive_Control_Miseq.paths.json; *Saccharomyces* (4930) Control TASS 98.51, 386 reads, source: Positive_Control_Miseq.paths.json

Sample 2: Nanopore_Run_A

SAMPLE TYPE	SEQUENCING PLATFORM	TOTAL READS	ALIGNED READS	SPECIES GROUPS	TASS CUTOFF
stool	oxford	3,000	2,541	9	68.23

	Organism	TASS Score	Classifier Reads	Aligned Reads	RPM	% Coverage
	Listeria *1 species; 1 qualifying strain*					
	Listeria monocytogenes EGD-e ★ ■ (169963) *strain*	99.8	199	199 (6.6%)	78,316	29.6%
	Pseudomonas *1 species; 1 qualifying strain*					
	Pseudomonas aeruginosa ■ (287) *species*	100.0	8	9 (0.3%)	3,148	1.6%
	Salmonella *1 species; 1 qualifying strain*					
	Salmonella enterica ★ ■ (28901) *species*	100.0	42	96 (3.2%)	17,005	97.9%
	Escherichia *1 species; 1 qualifying strain*					
	Escherichia coli O157:H7 ★ ■ (83334) *strain*	71.1	20	20 (0.7%)	7,871	1.9%

Sample 3: Negative_Control_Miseq

SAMPLE TYPE	SEQUENCING PLATFORM	TOTAL READS	ALIGNED READS	SPECIES GROUPS	TASS CUTOFF
nasal	illumina	172	144	10	75.0

No organisms met the reporting confidence threshold for this sample.
3 high-consequence detections were identified below the threshold. See the Low Confidence, High Consequence Detections section for details.

Sample 4: Positive_Control_Miseq

SAMPLE TYPE	SEQUENCING PLATFORM	TOTAL READS	ALIGNED READS	SPECIES GROUPS	TASS CUTOFF
nasal	illumina	38,000	33,614	12	75.0

	Organism	TASS Score	Classifier Reads	Aligned Reads	RPM	% Coverage
	Orthopneumovirus *1 species; 1 qualifying strain*					
	human respiratory syncytial virus (B) ■ (11250) *strain*	99.0	25	50 (0.1%)	1,487	56.7%
	Neisseria *1 species; 1 qualifying strain*					
	Neisseria gonorrhoeae ■ (485) *species*	99.1	3,493	6,986 (18.4%)	197,745	61.5%
	Orthopoxvirus *1 species; 1 qualifying strain*					

	Organism	TASS Score	Classifier Reads	Aligned Reads	RPM	% Coverage
	Monkeypox virus ★ ■ **(10244)** ***strain***	80.4	4	8 (0.0%)	238	1.2%
	Salmonella ***1 species; 1 qualifying strain***					
	Salmonella enterica ★ ■ **(28901)** ***species***	87.2	79	193 (0.5%)	5,059	1.1%
	Saccharomyces ***1 species; 1 qualifying strain***					
	Saccharomyces cerevisiae ■ **(4932)** ***species*** **! Saccharomyces cerevisiae S288C** **TASS 51.7 368 reads**	98.5	193	386 (1.0%)	832	1.4%

Low Confidence, High Consequence Detections

Detected strains below their sample-specific confidence thresholds.

	Sample	Organism	TASS Score	Classifier Reads	Aligned Reads
	Negative_Control_Miseq	Listeria monocytogenes (1639) ★	19.6	4	6
	Negative_Control_Miseq	Salmonella enterica (28901) ★	21.0	7	10
	Positive_Control_Miseq	Listeria monocytogenes (1639) ★	36.3	49	83

Additional Information

• Sterile samples (**blood, CSF, sterile, and serum**) may yield lower TASS scores due to relatively low read count or limited genomic coverage. All detected pathogens for these sample types are therefore classified as **primary pathogens** by default.

• Sample names and species groups throughout the report are hyperlinked for navigation. Selecting a sample or organism entry will jump directly to its corresponding section in the document.

• Organism groups are sorted by TASS score by default, or alphabetically when the alphabetical sorting option is enabled.

• Only samples and organism groups containing visible qualifying strains are included in the index and navigation sections.

• When subkey grouping is enabled, each genus-level group expands into species/subkey summary rows followed by qualifying child strains that also pass the reporting threshold.

• Low-confidence, high-consequence detections that do not appear in the PDF report may still be present in the corresponding Discovery Analysis TXT output file.

• Please visit our **DOCUMENTATION PAGE** for additional details on TASS confidence scoring.

Questions, feedback, or issues related to this report or the TaxTriage workflow can be submitted through the GitHub repository. General questions and discussions should be posted in the **Discussions** section, while bugs and feature requests should be submitted through the **Issues** tracker.

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Samples in Run:

Format: Sample name (# Total Alignments;# Primary Pathogens) → Category Label ■ (# Primary Strains, Max TASS)